# Health-a-thon 2026

FINAL SUBMISSION

# Sahara

A phone app for people with a serious illness, like advanced cancer, and their families. Their doctor discusses their end of life wishes; Sahara creates a clinical document, and an emergency doctor can read it in one scan.

**TEAM NAME** Zeros and Ones **TEAM CODE** 37MYUP

**Shaik Abdul Basith · Shaik Muhammad Awaiz · Dr Sharada Vinod Kutty · Dr Sujay Halkur Shankar · Shirin Ayub**

Try it: sahara.wedevit.in · Android app and website · English, Hindi, Telugu, Urdu, Kannada

# Team Overview & Background



### Shaik Abdul Basith

**Technical lead · Founder**

**WeDevIT Private Limited · wedevit.in**

Led the build end to end: the website and the Android app; the Gemini pipeline that transcribes the conversation, fills the goals-of-care document and answers questions from reports; the Supabase server, security and Google Play releases; the hospital survey site; this deck and brief



### Shaik Muhammad Awaiz

**Android engineer**

**WeDevIT**

Built Android screens and flows: patient summary, side-effect notes, care timeline, doctor schedule, sign-up and family invites by link, the Today, Care and More redesign, welcome panel and screen transitions; AI retry and backup models; tested every version



### Dr Sharada Vinod Kutty

**Pulmonologist**

**AIIMS Mangalagiri**

Wrote Saanthvana v1.0, the goals-of-care structure the app grew from, then the clinical specification and platform spec (29 Oct) and the comfort-care rule. Designed the goals of care document; set how often patients log symptoms; reviewed the symptom list; tested the survey and shared it with colleagues; advised on medicolegal limits of AI



### Dr Sujay Halkur Shankar

**Pulmonologist**

**Tata IISc Medical School**

Defined the problem statement, evidence and solution, and wrote the final problem, solution and features text (7 Nov); distilled the goals-of-care form; designed and circulated the hospital survey; positioned Sahara as the patient-facing extension of CARE; arranged legal advice; tested the whole app (3 Nov); designed the 13-topic checklist and guided conversation (7 Nov); led the clinical wording review



### Shirin Ayub

**Medical student**

Designed the symptom log: the symptom list, how often to log, and the trend graph for the doctor. Defined caregiver portal needs and researched caregiver stress for the wellbeing check. Reworked the patient story in our first deck; asked hospitals about morphine access; UI clean up and usability inputs

**Our doctors** chose this problem from their own clinics, wrote the clinical rules and checked every version.

**Legal advice was sought:** inputs and refining the goals of care wording and format.

Built by WeDevIT, the team's company. The app is on Google Play under WeDevIT.

# Problem Statement and Supporting Evidence




*A typical case: a 62-year-old woman with advanced breast cancer.*

**At the clinic,** her doctor asks what treatment she would want if she got very sick. Only her husband is there.

**Two months later** she is unconscious in the emergency room. Should doctors try to restart her heart (CPR)?

**Her children disagree,** and nobody can find what she said.

**98%**

of India's metastatic (advanced) cancer patients had unmet palliative-care needs.

Patil, BMJ Support Palliat Care 2024

**30–37%**

of patients get timely palliative care referrals (two centres).

Venketeswaran, BMJ Support Palliat Care 2026

**41%**

of cancer patients dying on active treatment had end-of-life care documented (one hospital).

Das, Cancer Med 2026

**88%**

of urban adults want to choose their own end-of-life care.

Living Will Survey, IndiaSpend 2019

### What has to change

**Her goals of care need to be documented, signed, and found in seconds in an emergency.**

# Target Users and Use Case



**Patient** · at home
Logs symptoms, medications, and helps find care; access to clinical documents

**Family** · at home
Access to their patient's portal; logs specific caregiver issues

**Their doctor** · clinic
Guides goals of care discussions; access to patient information: symptom trends, reports, medications

**Emergency doctor** · emergency room
Quick retrieval of the latest goals of care discussion to guide emergency medical decisions

## Words used in this deck

**Palliative care:** care that eases pain and distress in a serious illness.

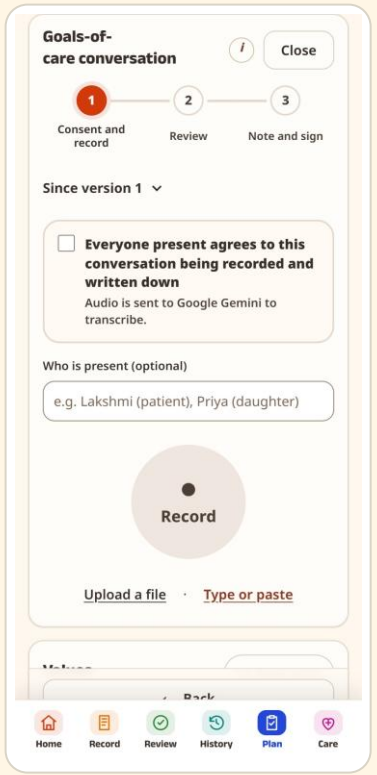
**Goals of care:** what treatment a patient wants, or refuses, near the end of life (CPR, ventilator, ICU).

**QR code:** a square barcode on the patient's phone or card; a phone camera scan opens their wishes.

# Solution Overview, Key Features & Impact (1/2)

## In the clinic: the doctor talks, Sahara listens and writes.

After consent, the doctor leads the goals-of-care conversation and records it on the doctor portal. Today this is rarely documented.



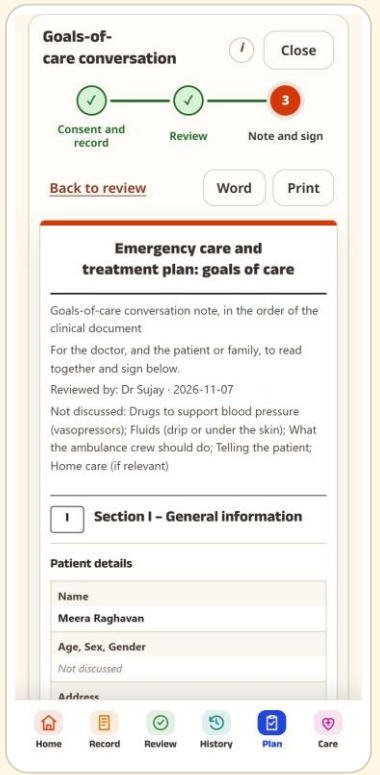

### 1. Patient agrees, tap Record

Google's Gemini AI types out who said what



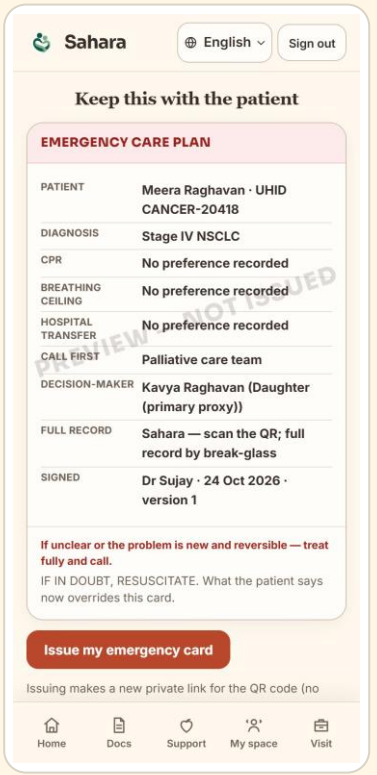

### 2. A checklist ticks itself

Missed topics are flagged. Android: Dr Sujay's 13 topics, with "Guide me" or "Free flow" taps




### 3. Answers fill the structured clinical document

Doctor checks the transcription and summary



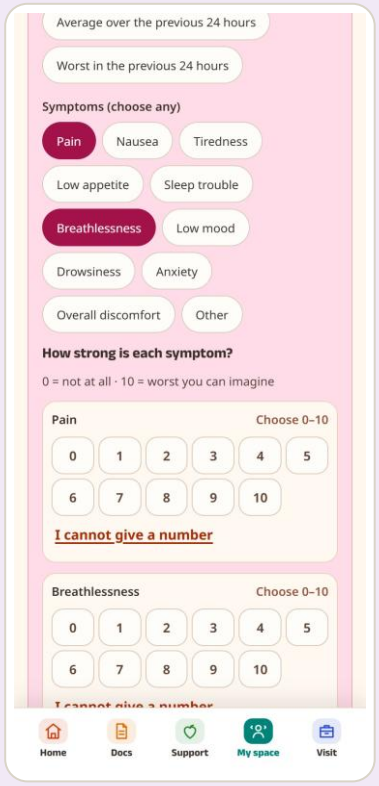

### 4. Everyone signs on the phone

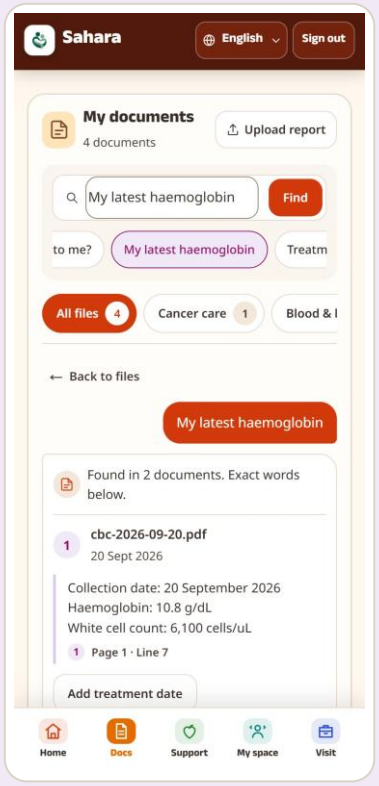
Doctor and patient or family sign; it locks and opens from a QR code in an emergency

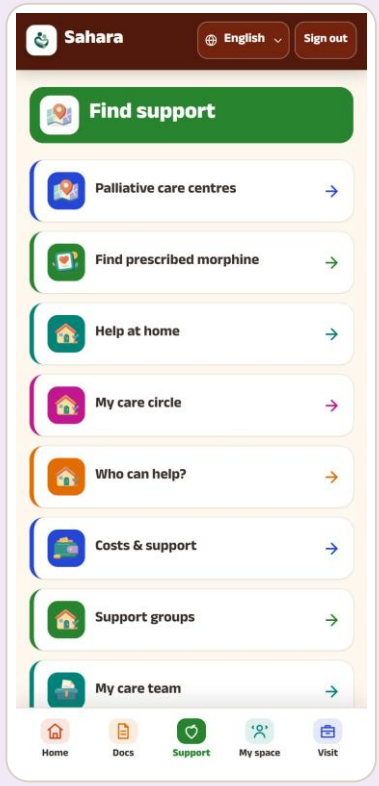
# Solution Overview, Key Features & Impact (2/2)

**At home, between visits:** patients and families log how things are going; their doctor sees the trends.

*Every feature is built on published guideline tools and validated questionnaires (ESAS-r symptoms, NCCN Distress Thermometer, ZBI-4 caregiver burden).*




- **How I feel:** symptoms (ESAS-r), a 0–10 distress scale (NCCN thermometer) and a pain body map; trends for the doctor.
- **Caregiver wellbeing (new):** a short check for family; flagged with Tele-MANAS 14416 and support.
- **Medicines:** pill tracker from prescription photos; tag pictures of your medicine strips for easy identification; dose ticks, reminders.
- **Documents:** uploads sorted into prescriptions, bills, reports; questions answered with the line it came from.
- **Emergency:** SOS button and a QR code with goals of care, contacts and medicines.
- **Help nearby:** palliative-care centres and availability of morphine, emergency hospitals, ambulances, support groups.
- **Family:** invited relatives share the record with their own access level; ability to manage caregiver concerns.

**Impact we will measure in the clinic pilot:**

more of these talks written and signed · wishes found in seconds in an emergency · fewer family arguments at the bedside · doctors see problems before the next visit.

# Technology, Tools, and AI Approach

## How a conversation becomes a signed clinical document

**Doctor's phone**
records only after patient consent

→

**Our server**
keeps the AI password; the phone never has it

→

**Google Gemini AI**
types out the talk and fills in the clinical document

→

**Doctor**
reads it, corrects it, signs; the patient or caregiver also signs

→

**Locked document**
PDF + QR code; any later change creates a new version that requires re-signatures

**No internet, or the AI is busy?** OpenAI Whisper turns speech into text on the phone itself. If that fails too, the doctor types.

## Built with

**Android**
the phone app

**Kotlin + Compose**
language and screens of the app

**Next.js**
the website

**Vercel**
puts the website online

**Supabase**
server, sign-in, storage

**PostgreSQL**
database; each family sees only its own

**Google Gemini**
AI: speech to text, fills the document

**OpenAI Whisper**
speech to text with no internet

**Google ML Kit**
5 languages and read-aloud on the phone

**OpenStreetMap**
free map of nearby centres

**Clinical tools**
ESAS-r, NCCN Distress Thermometer, ZBI-4

**FHIR**
standard format hospital software can read

# Build Sprint Journey and Progress

## October – November 2026

Built · Doctor / legal feedback · Milestone

| | MON | TUE | WED | THU | FRI | SAT | SUN |
|---|---|---|---|---|---|---|---|
| Week 1 | 5 Oct | 6 | 7 | 8 | 9<br>Shortlisted!<br>Android app + map of centres | 10<br>AI medicine organiser | 11<br>Prescription organiser |
| Week 2 | 12 | 13<br>Google Play test version | 14<br>Testers invited | 15 | 16<br>Privacy and account safety | 17 | 18 |
| Week 3 | 19<br>Sent to Google Play | 20<br>Mentor review | 21 | 22 | 23<br>Hospital survey feedback | 24 | 25<br>Symptom tracker |
| Week 4 | 26<br>Pain body map | 27<br>Hospital survey live | 28<br>Team meeting: 9 changes<br>Ask your reports | 29<br>Dr Sharada's medical rules | 30<br>Goals of care + QR code<br>Telugu, Urdu, Kannada | 31<br>Legal advice on format | 1 Nov<br>Recording → text |
| | 2<br>Doctor schedule | 3<br>Dr Sujay tests the whole app | 4<br>Clinical document word for word<br>8 practice talks pass | 5<br>Gemini AI + backups | 6<br>Calmer, fewer taps | 7<br>Dr Sujay's 13 topics<br>Caregiver wellbeing | 8 |

*Blue circle: 7 Nov, the day of this submission. Grey days are outside the build sprint.*

### ✓ Working now

Patient and family app · doctor's view · recording → signed clinical document · QR code · website live

### ↻ Changed on the way

Website first, then a real Android app. The AI used to write the note; now the doctor leads and the AI only writes it down.

### ⚠ Problems we solved

Google's AI was often busy → automatic backups. Exact goals-of-care wording → copied.

### → Next steps

Field testing on more phones · recorded practice talk with Dr Sharada · clinic pilot

# Evaluation and User Validation

## 1. Does it work? Our own tests

**877**
automatic checks on the Android app: all pass

**594 / 594**
automatic checks on the website: all pass

**8 / 8**
practice doctor–patient talks in 6 languages: every answer landed in the right place on the document

**4 / 4**
questions about a sample hospital report answered correctly

*Automatic checks are small test programs that run after every change, so a broken feature is caught before release. Practice talks use scripted conversations, so no patient data is exposed.*

## 2. What experts told us, and what we changed

**Dr Sujay Halkur Shankar**
doctor, used the whole app on 3 Nov

**Said:** Is there a guide telling the doctor what to talk about next? And a medicine taken 4 times a day needs 4 ticks.

**We changed:** Checklist in the doctors' order with a "not yet discussed" prompt; one tick per dose. Team chat: tubes-and-drains tracker, darker text.

**Dr Sharada Vinod Kutty**
doctor, wrote our medical rules on 29 Oct

**Said:** Comfort care must always continue, whatever treatment a patient refuses.

**We changed:** The app shows this on every signed document.

**Next steps:** clinic pilot with patients · field testing on more phones · speech-to-text on Hindi, Telugu and Kannada clinic recordings · doctor review of every translation

# Safety and Responsible AI

## Privacy and consent

- ✓ Recording starts only after the patient ticks “I agree”
- ✓ The AI is used only with that consent
- ✓ Each family sees only their own records

## Limits on the AI

- ✓ The AI never suggests treatment
- ✓ If a topic was not discussed, it says so; it does not guess
- ✓ If the AI is down, the phone or the doctor writes it

## A person decides

- ✓ The doctor reads, corrects and signs first
- ✓ The patient or family signs too
- ✓ Any later change creates a new version requiring re-signatures

## For everyone

- ✓ 5 Indian languages, and it reads text aloud
- ✓ Sign with a thumbprint; large text
- ✓ Demos use sample patients, so no real data is exposed

# Deployment, Cost and Sustainability Plan

### Pilot
Start in 1 clinic with our own doctors. Patients give consent as India's data-protection law (DPDP Act) requires.

### Fitting in
Today it records on the doctor's phone. Next: ambient listening, with room or lapel microphones in clinics. Later, FHIR export for hospital software.

### Who pays
Clinics or charities pay a monthly fee. Free for patients and families.

### Growing
Grants first, then palliative-care networks across India.

| What it costs | Once | Every month |
|---|---|---|
| Servers and website (Supabase + Vercel) | – | ₹4,200 |
| Google Gemini AI, per 30-minute talk | – | ≈ ₹2 a talk |
| Setting up and training clinic staff | ₹25,000 | – |
| Consent forms and legal check | ₹25,000 | – |
| Support from our team | – | ₹10,000 |
| **Cost per clinic, if 10 clinics share it** | **₹5,000** | **≈ ₹1,600** |

**We would charge ₹1,500–2,000 per clinic a month**

*Based on public prices: US$1 ≈ ₹94; Supabase $25 and Vercel $20 a month; Gemini ≈ $0.30 per million audio units (a 30-minute talk ≈ 58,000); 100 talks per clinic a month.*

# What Next? Plans to take initiative forward (if any)

1

## Field testing

Test on more phones; try room and lapel microphones (ambient listening). Record a real practice talk with Dr Sharada.

2

## Pilot, 60–90 days

One clinic. Count how many talks get signed and how fast wishes are found.

3

## Legal rules

What to do when a patient cannot decide, who may sign for them, and how a patient changes their goals of care.

4

## Fund and grow

Grants, palliative-care networks, then link with hospital software.

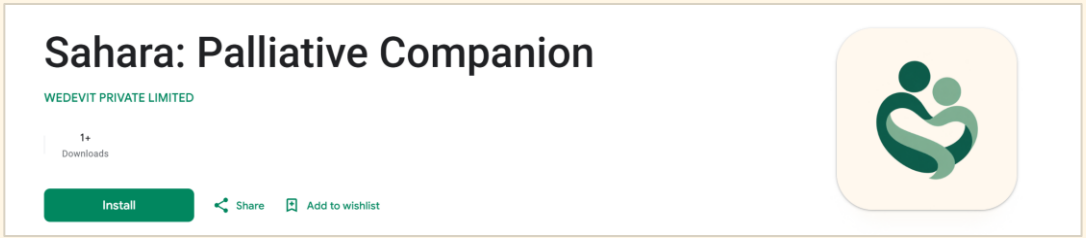
**Next version of the app:** more features such as devices like feeding tubes · an improved caregiver portal · updates based on user feedback

**Can the organisers share our solution publicly?** YES

# Health-a-thon 2026

# Thank you

**Zeros and Ones • 37MYUP**




**Android app on Google Play**

**Website: sahara.wedevit.in**

Try it: sahara.wedevit.in | Demo video: [add demo video link] | workwithdevit@gmail.com | Code: github.com/basithladdu/healthathon (private)

**Shaik Abdul Basith · Shaik Muhammad Awaiz · Dr Sharada Vinod Kutty · Dr Sujay Halkur Shankar · Shirin Ayub**

**Health-a-thon 2026**